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1. Administrative & Study Identification

Full Study Title: A Phase IV Open-Label Study Evaluating the Effectiveness and Safety of Risdiplam Administered in Pediatric Patients With Spinal Muscular Atrophy Who Experienced a Plateau or Decline in Function After Gene Therapy **Short Title/Acronym:** HINALEA 2 **Protocol Number:** BN44621 / 2023-505161-81-00 **NCT Number:** NCT05861999 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Risdiplam (Evrysdi) **Phase of Development:** PHASE4 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effectiveness and safety of risdiplam administered in pediatric participants with Spinal Muscular Atrophy (SMA) and 2 SMN2 copies who previously received onasemnogene abeparvovec (gene therapy) and experienced a plateau or decline in function. **Secondary Objectives:** To assess changes in swallowing function, motor development, respiratory function, physical growth, and the safety profile of the study treatment over time.

2.2 Background & Rationale Currently, there are reported cases of risdiplam being started after SMA gene therapy (onasemnogene abeparvovec), yet there is no standard clinical pathway for this combination. Some physicians initiate risdiplam early after gene therapy, while others start it later when the patient may not be responding well. The HINALEA 2 trial is designed to address this gap by systematically evaluating the use of risdiplam as a supplementary, non-invasive, disease-modifying therapy for infants and toddlers under 2 years old who demonstrate a plateau or decline in function post-gene therapy.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Single-arm)

3.2 Planned Enrollment & Duration Trial Status: RECRUITING **Target N:** 28 participants **Number of Sites:** Multicenter **Estimated Study Start:** 14-Aug-2024 **Estimated Primary Completion:** 11-Feb-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. <2 years of age at the time of informed consent. 2. Confirmed diagnosis of 5q-autosomal recessive SMA, including genetic confirmation of homozygous deletion or compound heterozygosity predictive of loss of function of the Survival of Motor Neuron 1 (SMN1) gene. 3. Confirmed presence of two SMN2 gene copies as documented through laboratory testing. 4. Administration of onasemnogene abeparvovec pre-symptomatically or post-symptomatically. 5. Has received onasemnogene abeparvovec for SMA no less than 13 weeks prior to enrollment. 6. If treated with risdiplam prior to onasemnogene abeparvovec, risdiplam treatment must not have exceeded 3 weeks and must be discontinued 1 day prior to onasemnogene abeparvovec administration. 7. In the opinion of the investigator, has demonstrated a plateau or decline in function post-gene therapy (with a duration of 26 weeks or less) documented by 2 individual time points in the functions as follows: swallowing AND one additional function/ability (respiratory, motor function, other) per appropriate expectation.

4.2 Exclusion Criteria 1. Previous or current enrolment in investigational study prior to initiation of study treatment. 2. Any unresolved standard-of-care laboratory abnormalities per the onasemnogene abeparvovec prescribing information. 3. Concomitant or previous administration of an SMN2-targeting antisense oligonucleotide. 4. Concomitant or previous use of an anti-myostatin agent. 5. Participants requiring invasive ventilation or tracheostomy. 6. Presence of feeding tube and an OrSAT score of 0. 7. Hospitalization for pulmonary event within the last 2 months, or any planned hospitalization at the time of screening. 8. Any major illness requiring hospitalization within 1 month before the screening examination or any febrile illness within 1 week prior to screening and up to first dose administration.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Risdiplam administered orally once daily. **Route of Administration:** Oral **Duration of Treatment:** 72 weeks (Treatment Period), followed by a 1-year Treatment Extension Period for a total study duration of 120 weeks (approximately 2.5 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for SMA including respiratory and nutritional support. **Prohibited:** Concomitant use of other investigational SMA therapies or additional doses of gene therapy.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Genotype Confirmation, Baseline Motor Assessment
Baseline	Day 0	First Dose, Caregiver Administration Education
Interim	Every 12 Weeks	Safety Labs, BSID-III Motor Assessment, Swallowing/Respiratory Evaluation, IP Accountability
End of Treatment	Week 72	Primary Endpoint Assessment (Gross Motor Score)
End of Study	Week 120	Final Vital Signs, Exit Interview, Conclusion of 1-Year Extension

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from Baseline in the Raw Score of Bayley Scales of Infant and Toddler Development - Third Edition (BSID-III) Gross Motor Score at 72 Weeks of Risdiplam Treatment. The BSID-III is a standardized assessment commonly used to evaluate developmental functioning of infants and young children between 1 month and 42 months of age. The gross motor scale measures the movement of the limbs and torso. Items assess static positioning (e.g., sitting, standing); dynamic movement, including locomotion and coordination; balance; and motor planning. The gross motor scale consists of 72 items scored at 0 (unable to perform) or 1 (criteria for item achieved). A higher raw score indicates improvement. **Measurement Method:** Bayley Scales of Infant and Toddler Development, Third Edition (BSID-III).

7.2 Secondary & Exploratory Endpoints - Percentage of Participants With Adverse Events. - Percentage of Participants With Serious Adverse Events. - Percentage of Participants With Treatment Discontinuation Due to Adverse Events. - Changes in swallowing function assessed at 72 weeks and over time. - Changes in respiratory function. - Changes in overall motor development. - Measurement of child growth trajectories. - Safety profile and adverse events associated with study treatment.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by caregivers with formal assessments by the clinical trial doctor every 12 weeks. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the Sponsor. **Data Monitoring Committee (DMC):** Internal/External safety monitoring committee details managed by Hoffmann-La Roche.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy endpoints; Safety Set for all participants receiving at least one dose of risdiplam. **Sample Size Justification:** Target $N = 28$ participants, providing adequate clinical insight into the treatment trajectory of this specific pediatric rare disease sub-population. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) and Last Observation Carried Forward (LOCF) will be utilized for longitudinal motor scale data.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite monitoring and source data verification in alignment with the 12-week participant visit cadence. **Audit Trail:** Data entered into a validated EDC system with electronic signature controls and eCRF locking mechanisms.

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11. Study Identifiers & Governance

Primary ID: BN44621 **Registry Number:** NCT05861999 **Posting ID:** 502470000054311850 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** HINALEA 2 **Official Regulatory Title:** A Phase IV Open-Label Study Evaluating the Effectiveness and Safety of Risdiplam Administered in Pediatric Patients With Spinal Muscular Atrophy Who Experienced a Plateau or Decline in Function After Gene Therapy

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12. Clinical Framework (Spinal Muscular Atrophy Specific)

Primary Condition: Spinal Muscular Atrophy (SMA) **Disease Area:** Muscular Atrophy, Spinal **Definition:** The loss of muscle tissue due to inactivity or disease. **Semantic Type:** Pathologic Function **MeSH Code:** D009133 **SNOMED ID:** 88092000 **ATC Code:** M09AX10 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/evrysdi-epar-product-information_en.pdf **Diagnosis Threshold:** Confirmed 5q-autosomal recessive SMA, 2 SMN2 copies, age < 2 years **Medical Methodology:** Post-gene therapy (onasemnogene abeparvovec) rescue/additive treatment **Optimization Target:** Increasing and sustaining SMN protein production to improve gross motor and bulbar (swallowing) functions

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13. Study Procedures & Methodology

Management Pathway: Standardized introduction of once-daily oral risdiplam in pediatric patients experiencing functional plateau/ decline post-gene therapy. **Education Protocol (HCP):** Site training on standardized BSID-III administration and scoring protocols. **Education Protocol (Patient):** Caregiver education regarding daily oral administration (swallowed whole or dispersed in water) and handling of risdiplam at home. **Quality Audit Mechanism:** Centralized review and scoring alignment for BSID-III video assessments.

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14. Observation & Follow-Up Schedule

Total Duration: 120 Weeks (72 weeks treatment + 1-year extension) **Onsite Visit Cadence:** Every 12 weeks (Q12W) **Remote Monitoring Frequency:** As needed between scheduled 12-week clinical visits. **Checklist Review Interval:** Every 12 weeks (Q12W)

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |